HireWise AI

AI Recruitment System — Security Incident Scenarios

Scenario Background

HireWise is a company that uses an AI-powered recruitment system to screen CVs, rank applicants, and recommend candidates to HR staff. The system uses a large language model (LLM) to summarize CVs and generate candidate suitability scores.

Recently, the security team discovered suspicious activity involving the system. The following incident reports document the findings of the investigation.

Incident Reports

Incident 1 — Manipulated CVs

The HR team received several applications containing unusual instructions hidden in the CV text. When processed by HireWise, some of these CVs caused the AI assistant to ignore its normal instructions and produce unrelated text in its candidate summaries.

Incident 2 — Unauthorized Information Exposure

During testing, an HR employee asked the AI assistant to summarize a candidate's application. The assistant included internal notes about the candidate that were not supposed to appear in the summary. Investigators found that the notes were stored in the system's internal document database.

Incident 3 — Repeated API Queries

Security logs showed that an employee account submitted hundreds of slightly modified prompts to the AI system. The prompts asked for candidate rankings, scoring explanations, and examples of the model's responses. The account saved the outputs in local files.

Incident 4 — Suspicious External Model

Investigators discovered a personal project repository belonging to the employee. It contained the collected AI outputs and a script that used them to fine-tune a smaller language model. The project description stated that the goal was to reproduce HireWise's candidate-ranking behavior.

Incident 5 — Input Manipulation

The HR team also discovered that several CVs had been deliberately modified before being uploaded. Some contained misleading qualifications and instructions intended to influence the AI's ranking. However, investigators found no evidence that HireWise's underlying training dataset or model parameters had been modified.

Incident 6 — Unconfirmed Impact

The smaller model was shared with other users. Investigators confirmed that it had been trained using outputs from HireWise, but they could not establish whether it reproduced the original model's capabilities, exposed additional confidential information, or was used for further attacks.

Analysis Scope

This scenario is designed for threat modeling and technique mapping against the MITRE ATLAS (Adversarial Threat Landscape for Artificial-Intelligence Systems) framework. The analysis covers:

- Identification of relevant ATLAS tactics and techniques for each incident
- Evidence classification (confirmed vs. suspected)
- Attack chain reconstruction across incidents
- Recommended detection and mitigation strategies